

Loops and Vectorization

Giannacco

Sometimes, it is more efficient to have MATLAB perform calculations on an entire array rather than processing an array element by element. This can be done through vectorization.

For Loop

```
i = 0;  
for t = 0:0.02:50  
    i = i + 1  
    y(i) = cos(t);  
end
```

While Loop

```
i = 0;  
while (1)  
    i = i + 0.02  
    if i >= 50, break, end  
end
```

Vectorization

```
t = 0:0.02:50  
y = cos(t);
```